



System 6.1:

Ballute - Drogue



Overview



Burble Fence

Separates flow to create low pressure base drag



Side Inlet

Aids in inflation and drag

Apex Inlet

Uses rammed air to inflate the ballute

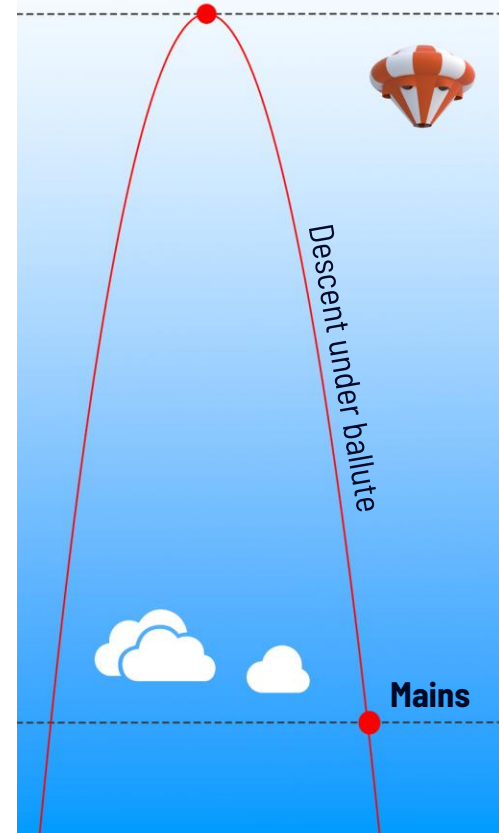
Ideal for **high-speed, low-density** decent

Increased low-pressure **stability**

Used in a **two-stage configuration** to minimise failure points

Tangling and **canopy collapse resistant** design compared to other solutions (DGBs)

Apogee, Ballute deployed

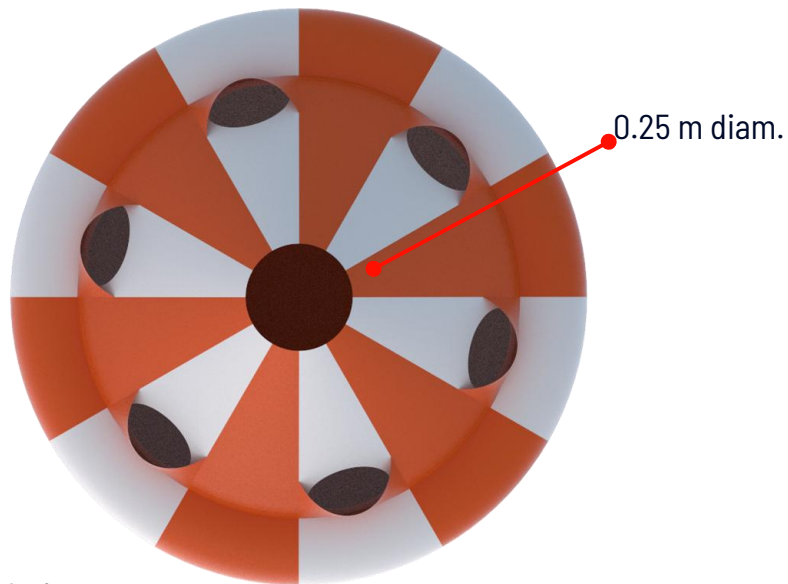
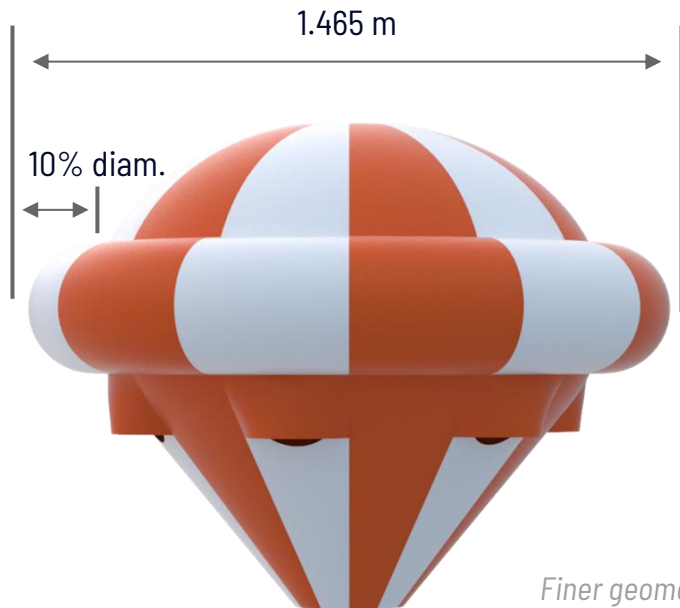


Sizing and Geometry

Diameter of 1.465 metres for a descent speed of **40 m/s** at **30,000 ft**

Resultant **24 m/s** descent speed **at mains**

★ Meets descent **requirements 5.3-1** and **5.3-2**



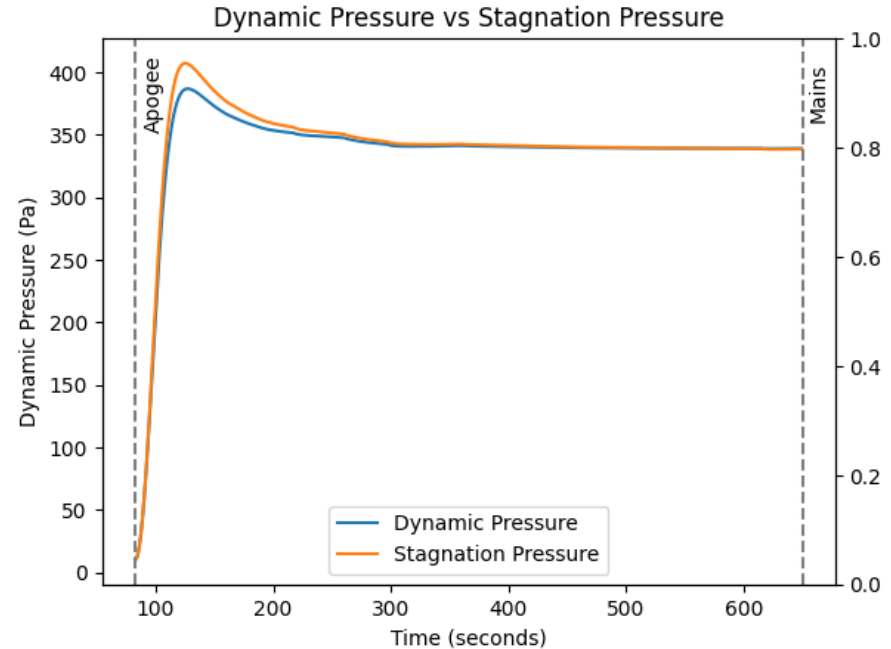
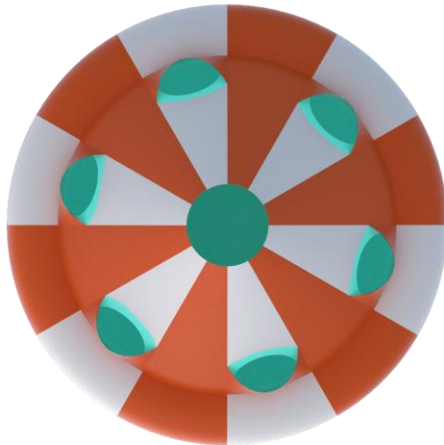
Finer geometry based on proven designs

Inlets and Inflation Pressure

The **stagnation pressure** of the airflow remains **above the stagnation pressure** for the **entire descent phase** under the ballute

The inlets were sized to a **minimum area of 566 cm²** for an **inflation time of 3 seconds** just after apogee.

Side inlets aid in **initial inflation** with **stiffened vents** in early deployment



Material Selection

Must withstand down to **-63 C** airflow
have **low porosity**

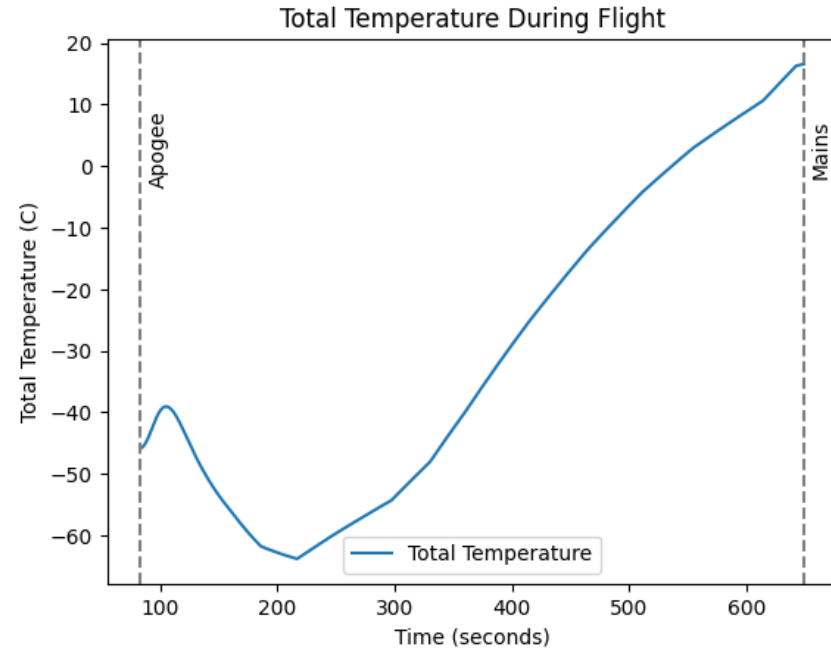
Must withstand **flagging forces** during inflation

Must

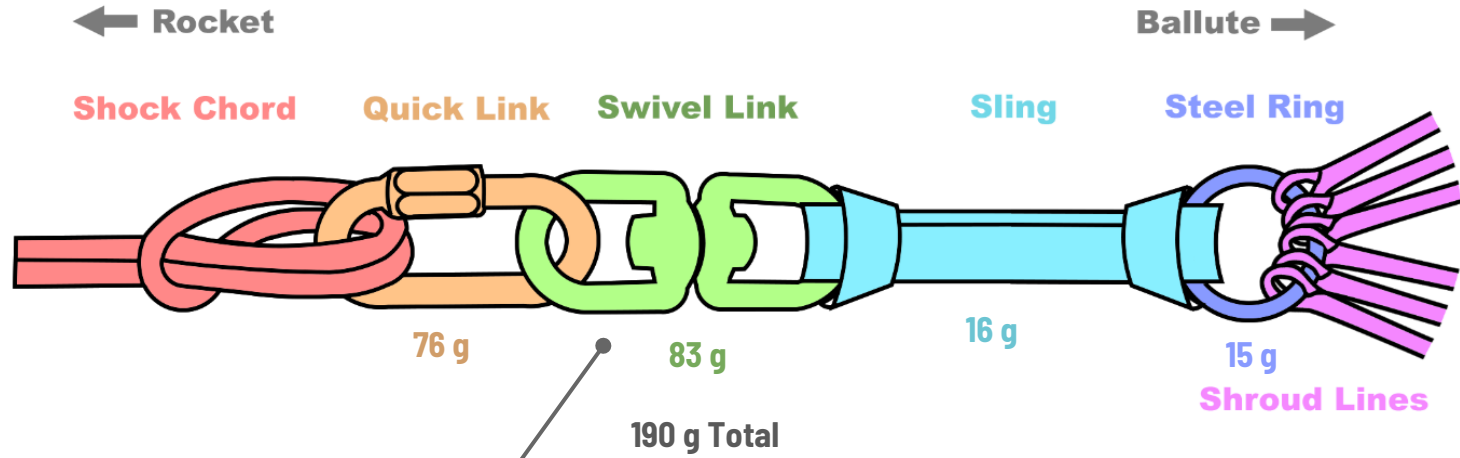
★ Chosen Material

50 D PU/Sil Coated **Ripstop Polyester**

- **High strength** in both weft and warp directions (min FoS of 5.9)
- **Zero porosity** through **polyurethane** and **silicon coating** (UV Resistant)
- Polyester, Polyurethane, and Silicon all have sufficiently low **Ductile-to-Brittle Transition Temperatures**.
- **Polyester fibre (PET)** has proved strength retention at extremely low temperatures.



Ballute Attachment



Separates ballute spin from main system

Maximum predicted **steady state descent** drag of **683 N**

Dyneema line **Long bury splice** for maximum shroud line strength retention



Manufacturing

Manufactured by MHPR to allow for **quick iteration** and **testing**

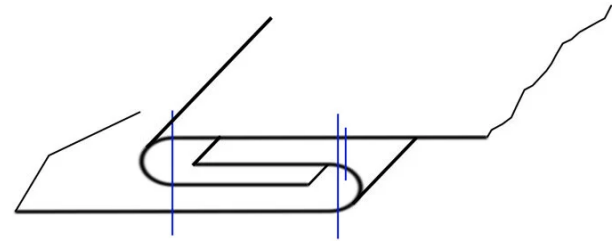
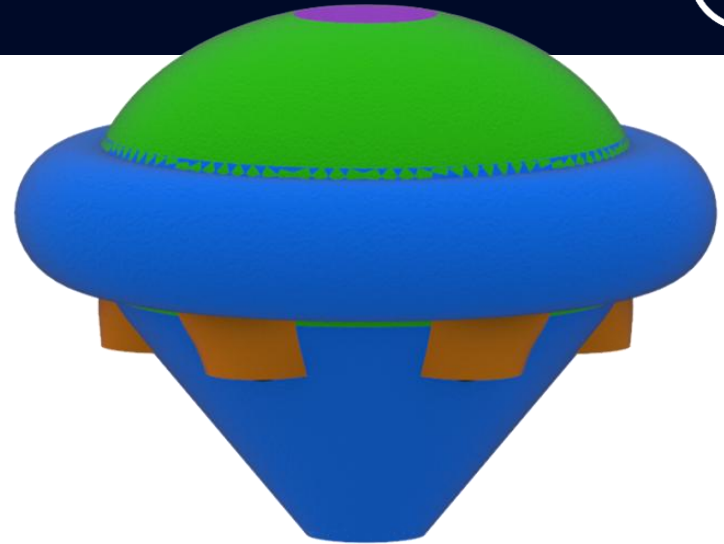
The ballute will be assembled in **four major stages**

The **upper half gores** and **lower half gores** are to be sewn together without completing the loop.

The **side inlet shrouds** will be sewn onto 6 of the 12 lower gores that will have inlet cutouts sewn in. The leading edges of these inlets will have **fabric boning integrated** to provide structure.

Both **upper** and **lower** halves will then be joined together inside-out, forming the burble fence along the upper edge.

All seams will be **flat felled seams** where possible with **double-stitched flat seams** elsewhere using **high strength polyester thread**.



Flat Felled Seam